4 **(a)** Describe an experiment to determine the specific heat capacity of aluminium. You may draw a diagram.

Include in your answer:

- the measurements made
- any equations needed.

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..... [4]

(b) An aluminium dish is initially at room temperature. Boiling water is poured into the aluminium dish as shown in Fig. 4.1.

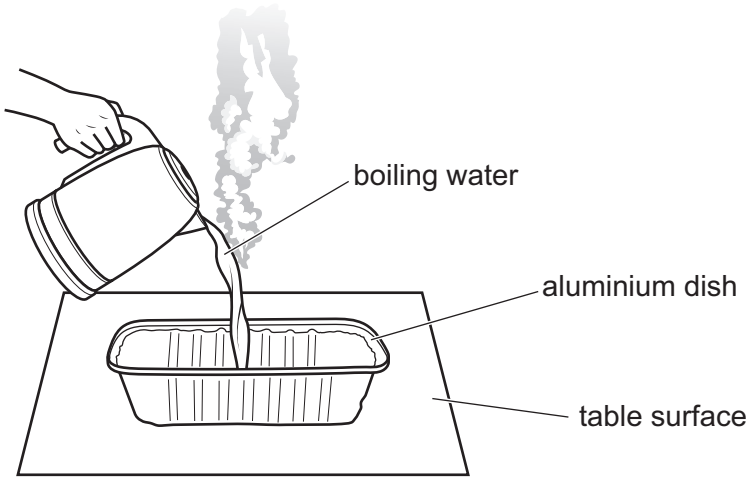

Fig. 4.1

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(i) Explain why, after a short time, the dish and the water are the same temperature.

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..... [3]

(ii) Explain, in terms of its particles, why the aluminium expands as the boiling water is poured into the dish.

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..... [2]

(iii) The water in the dish evaporates.

Explain, in terms of the water molecules, what is meant by evaporation.

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..... [2]

[Total: 11]

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